SOCIALMIND DRIVE

AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING



SLIIT
FACULTY OF COMPUTING

AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING



AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING

Users, Impact & Current Limitations

- Impact: Unsafe driving, congestion, accidents
- Limitations: Reactive systems, assume cooperation, ignore hidden intent, Cannot effectively handle right-of-way decisions at complex intersections

Proposed Solution

- Agentic AI framework
- Predict pedestrian intent In Occluded Scenarios
- Negotiate right-of-way between vehicles
- Understand human intention using Theory of Mind
- Collective learning between vehicles

Solution Overview & Advantages

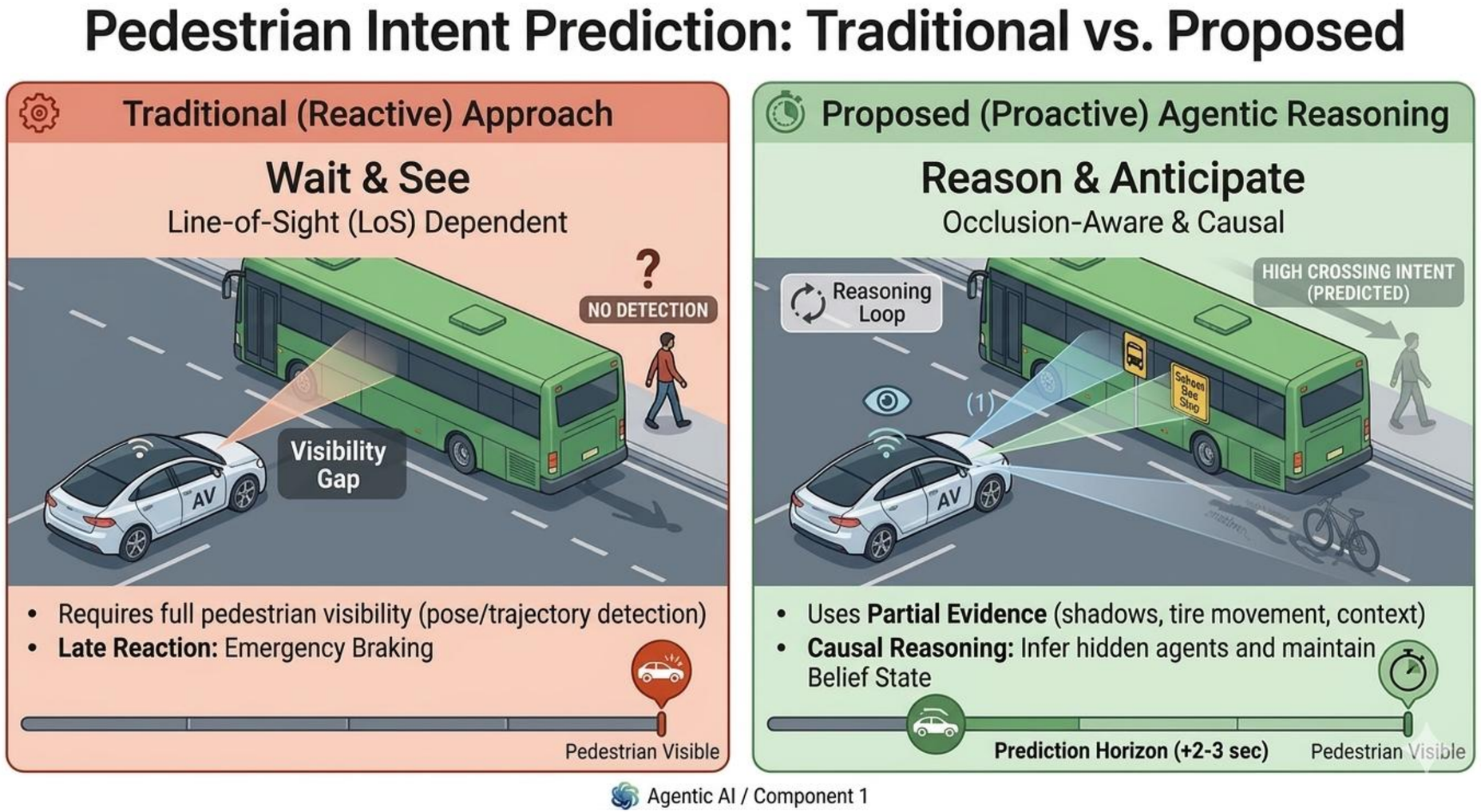
- Vehicles learn together, share knowledge selectively
- Anticipates pedestrian/cyclist actions
- Handles aggressive, distracted, or unpredictable road users
- Enables safe, natural, and socially aware driving
- Reduces accidents, congestion, and delays



IT22245960 | Bandara W A P S

Adaptive Intent Prediction in Occluded Urban Scenarios

Adaptive Intent Prediction in Occluded Urban Scenarios



Adaptive Intent Prediction in Occluded Urban Scenarios

1 **Problem Definition**

- Reactive Limitation
- Blind spots hide pedestrians, causing detection delays.
- Sudden "edge-case" appearances lead to high-risk collisions.
- Lack of Reasoning

2 **Novelty**

- Adaptive Reasoning
- Partial Evidence Integration
- Contextual Knowledge Fusion
- Causal Inference

Adaptive Intent Prediction in Occluded Urban Scenarios

✓ Comparison with Existing Systems

Single-Agent LLM Driving Systems	Multi-Agent LLM Driving Systems	Proposed System	Feature / Capability	Vision-Based Models	Pedestrian-Vehicle Interaction Models
✗	✗	✓	Predict intent before full visibility	✗	✗
✗	✗	✓	Works under occluded urban scenarios	✗	✗
✓	✓	✓	Uses pedestrian / cyclist interaction cues	✓	✓
✓	✓	✓	Real-time adaptive belief update	✗	✗

Adaptive Intent Prediction in Occluded Urban Scenarios

3 Research Objective

- To develop a robust Adaptive Intent Prediction framework for proactive autonomous navigation in occluded urban scenarios.

4 Sub-Objectives

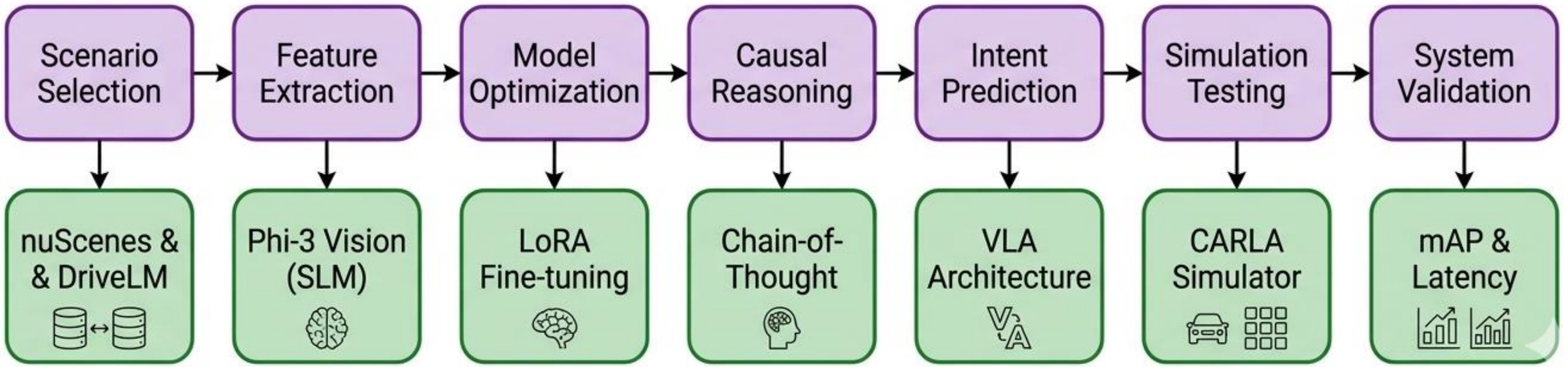
- Collecting Data
- Study intent patterns of road users.
- Develop an adaptive intent prediction framework.
- Support proactive decision-making in autonomous navigation.
- Test and validate the proposed framework.

Adaptive Intent Prediction in Occluded Urban Scenarios

✓ Application of Domain Knowledge

- Autonomous Vehicles
- Artificial Intelligence

✓ Research Methodology



Adaptive Intent Prediction in Occluded Urban Scenarios

Validation & Evaluation

- Simulation-Based Testing (CARLA Simulator)
- Qualitative evaluation by Domain Experts (AV researchers)
- Performance metrics

Adaptive Intent Prediction in Occluded Urban Scenarios

Data Availability

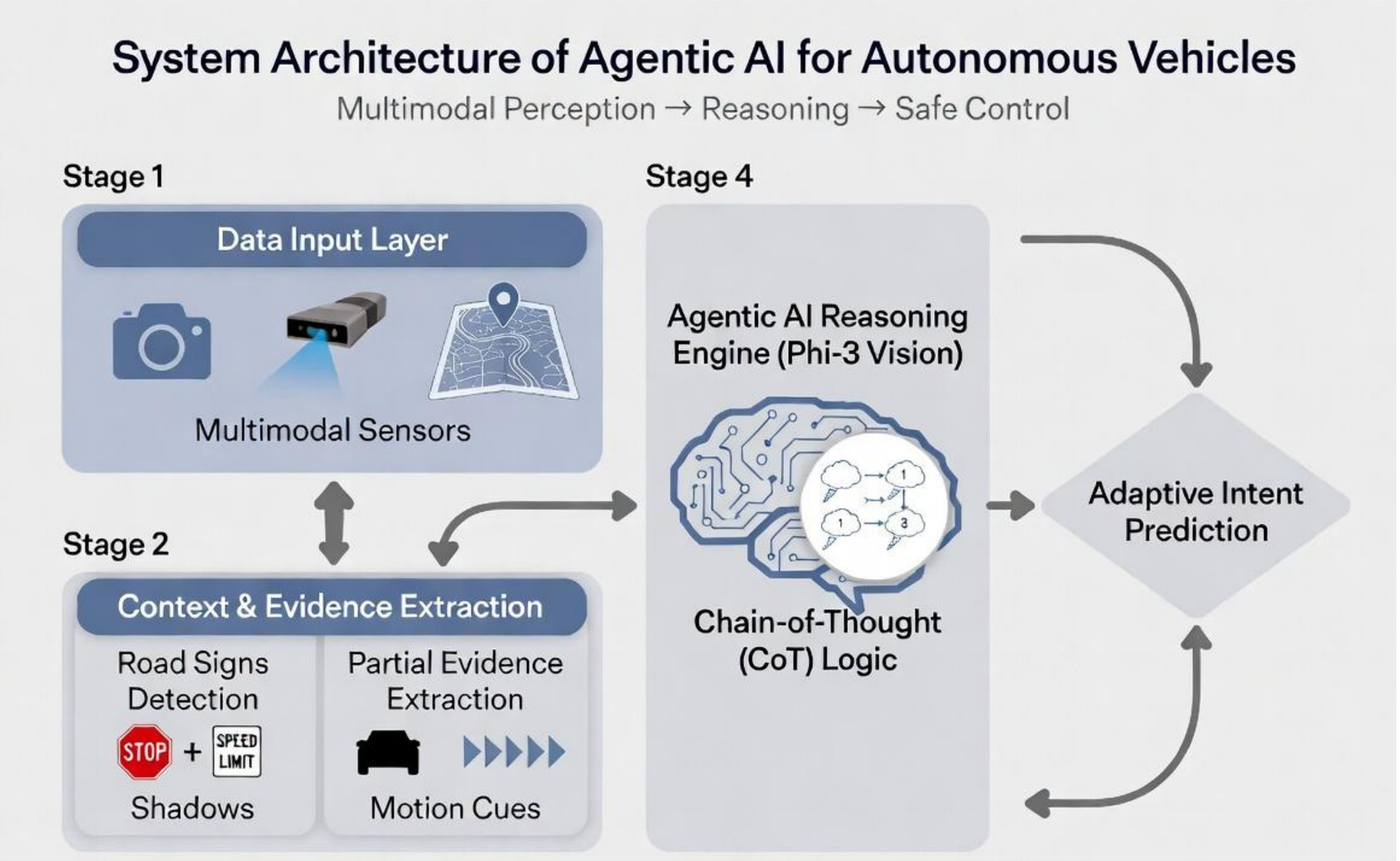
- Real-World Datasets: nuScenes , DriveLM
- Simulated Data: CARLA Simulator
- Multi-modal Input: High-resolution camera feeds

Security and Ethical Clearance

- Full anonymization of pedestrian and vehicle data
- Safety First

Adaptive Intent Prediction in Occluded Urban Scenarios

System Overview





IT22265906 | Nethmini R A I

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

Multi-Agent Negotiation for Right-of-Way in Complex Intersections



Multi-Agent Negotiation for Right-of-Way in Complex Intersections

1 Problem Definition

- AV crashes at intersections occur nearly twice as often as HDVs (68.5 % vs. 34.9 %) .

Limitations of Existing Systems

- Rule-based systems → cannot adapt to new situations
- Centralized systems → single point of failure
- MARL + rules → safety depends on predefined rules
- Game theory bidding → only one-time decision

2 Novelty

- Fully decentralized, continuous negotiation
- Safety emerges from learning, no external rules
- priority_intent signal (0-1) every 0.1s
- Real-time neighbour signal processing

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

✓ Comparison with Existing Systems

Feature / Capability	Rule-Based Systems	Centralized Systems	MARL + Rules	Game Theory Bidding	Proposed System
Adapt to new traffic situations	✗	✓	✓	✗	✓
No single point of failure	✓	✗	✓	✓	✓
Continuous negotiation between vehicles	✗	✗	✗	✗	✓
Fully decentralized decision making	✓	✗	✓	✓	✓
Safety learned automatically	✗	✗	✗	✗	✓

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

3 Research Objective

Develop a decentralized Multi-Agent Reinforcement Learning (MARL) framework where autonomous vehicles negotiate right-of-way at intersections to achieve:

- Zero collisions
- Improved traffic efficiency

4 Sub-Objectives

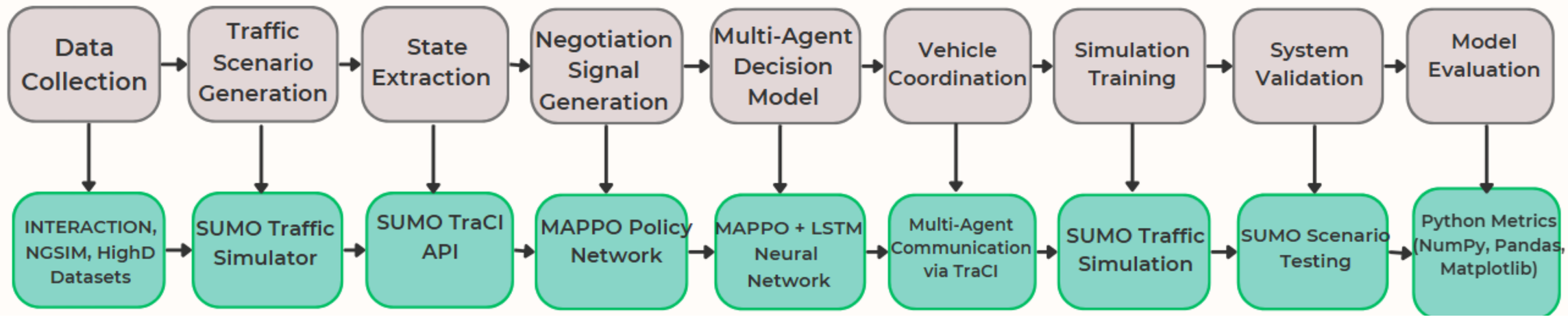
- Design communication framework (priority signals)
- Build learning model (MAPPO + LSTM)
- Train in simulation (SUMO & CARLA)
- Test vs existing methods (FCFS, MARL, rules, game-theory)

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

✓ Application of Domain Knowledge

- Autonomous Vehicles
- Artificial Intelligence
- Intelligent Transportation Systems

✓ Research Methodology



Multi-Agent Negotiation for Right-of-Way in Complex Intersections

✓ Validation & Evaluation

- Performance metrics: collision rate, travel time, throughput
- Safety metrics: post-encroachment time, deadlock frequency
- Fairness evaluation among vehicles
- Simulation-based testing (SUMO / CARLA)
- Qualitative evaluation by Domain Experts (AV researchers)

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

✓ Data Availability

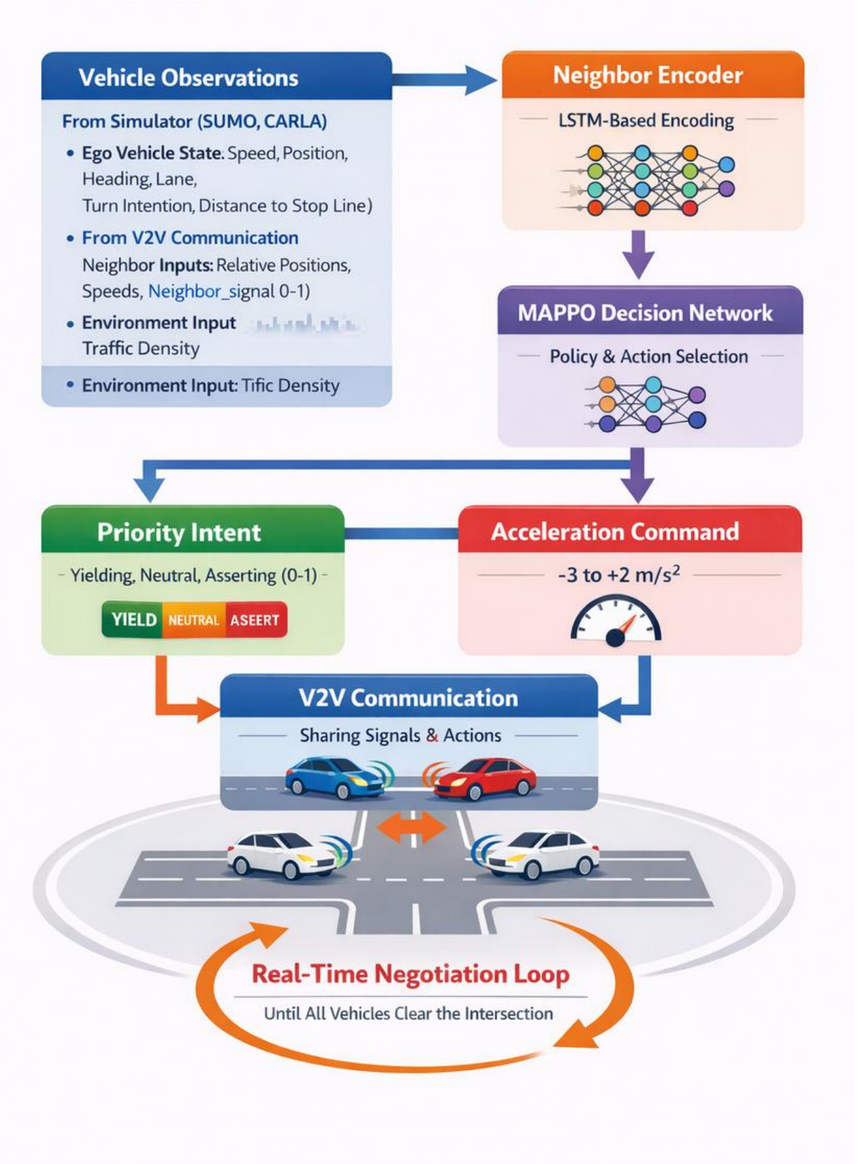
- Training: SUMO-generated synthetic traffic
- Validation: INTERACTION, NGSIM, HighD (real-world trajectories)
- Approach: Simulation for learning, real datasets for testing

✓ Ethical & Security Clearance

- Use public, anonymized datasets → no human participants
- Research only in simulation → no real-world risk
- Consider communication security risks

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

System Overview

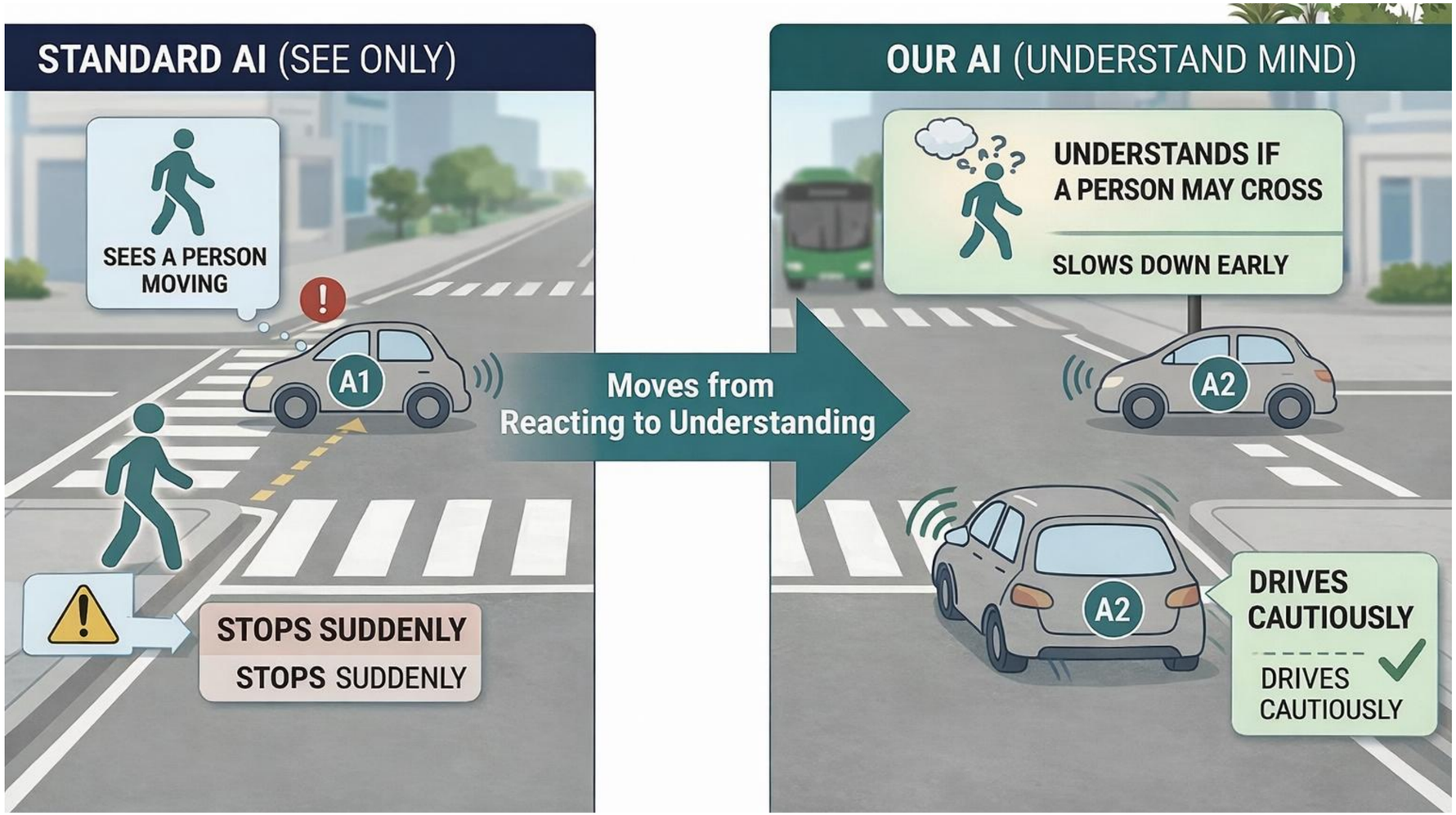




IT22223012 Withanage W W D T H

Proactive Social Compliance Modeling using Agentic Theory of Mind

Proactive Social Compliance Modeling using Agentic Theory of Mind



Proactive Social Compliance Modeling using Agentic Theory of Mind

1 Problem Definition

- Autonomous vehicles mostly react to observed pedestrian movements.
- Current models predict only trajectories or crossing/not-crossing.
- Uncertain behaviors like hesitation, distraction, or risky walking are ignored.and trafic norms are different accoding to culture.
- This can cause unsafe decisions in traffic.

2 Novelty

Theory-of-Mind Inspired Pedestrian Understanding

- Theory of Mind(ToM): ability to infer others' intentions or mental states.
- The system analyzes pose and motion cues to infer pedestrian mental states instead of only predicting crossing / not crossing.

Proactive and explainable AV actions

- The vehicle responds safely with human-readable reasoning based on pedestrian mental states and social norms.

Proactive Social Compliance Modeling using Agentic Theory of Mind

✓ Comparison with Existing Systems

Existing Work	Multi-State Mental Prediction	Cultural Adaptation
PedAST-GCN (Liu et al., 2024)	✗	✗
MonoPIC (Radhakrishna et al., 2024)	✗	✗
PedVLM (Munir et al., 2025)	✗	✗

Proactive Social Compliance Modeling using Agentic Theory of Mind

3 Research Objective

- To develop a Theory-of-Mind-inspired Agentic AI system for autonomous vehicles that infers pedestrian mental states and generates culturaly aware,proactive, explainable driving decisions.

4 Sub-Objectives

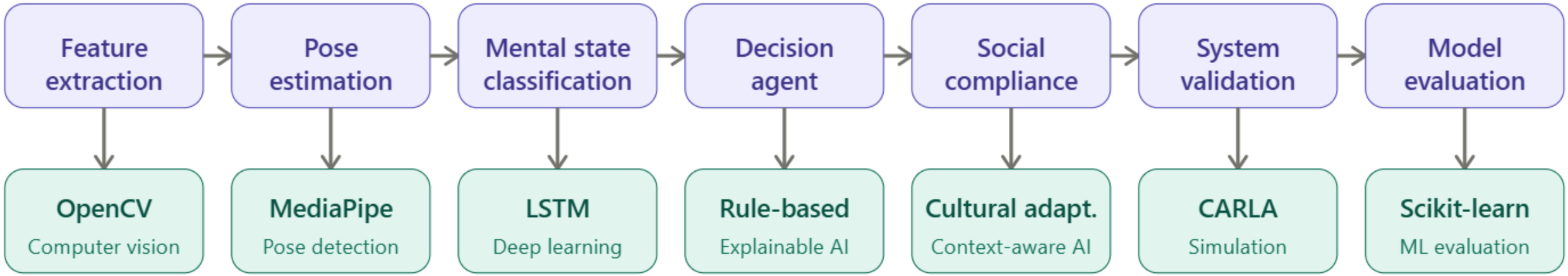
- Pedestrian data collection and feature extraction
- Pedestrian mental state classification
- Explainable rule-based decision agent development
- Social compliance modeling with cultural adaptation
- System validation using simulation environments

Proactive Social Compliance Modeling using Agentic Theory of Mind

✓ Application of Domain Knowledge

- Autonomous Vehicles
- Artificial Intelligence
- ToM Framework

✓ Research Methodology



Proactive Social Compliance Modeling using Agentic Theory of Mind

✓ Validation & Evaluation

- Accuracy, Precision, Recall, F1-score per mental state
- Confusion matrix analysis for 6 states
- Early prediction time measurement
- System Validation using Simulation environment
- Qualitative evaluation by Domain Experts (AV researchers)

Proactive Social Compliance Modeling using Agentic Theory of Mind

Data Availability

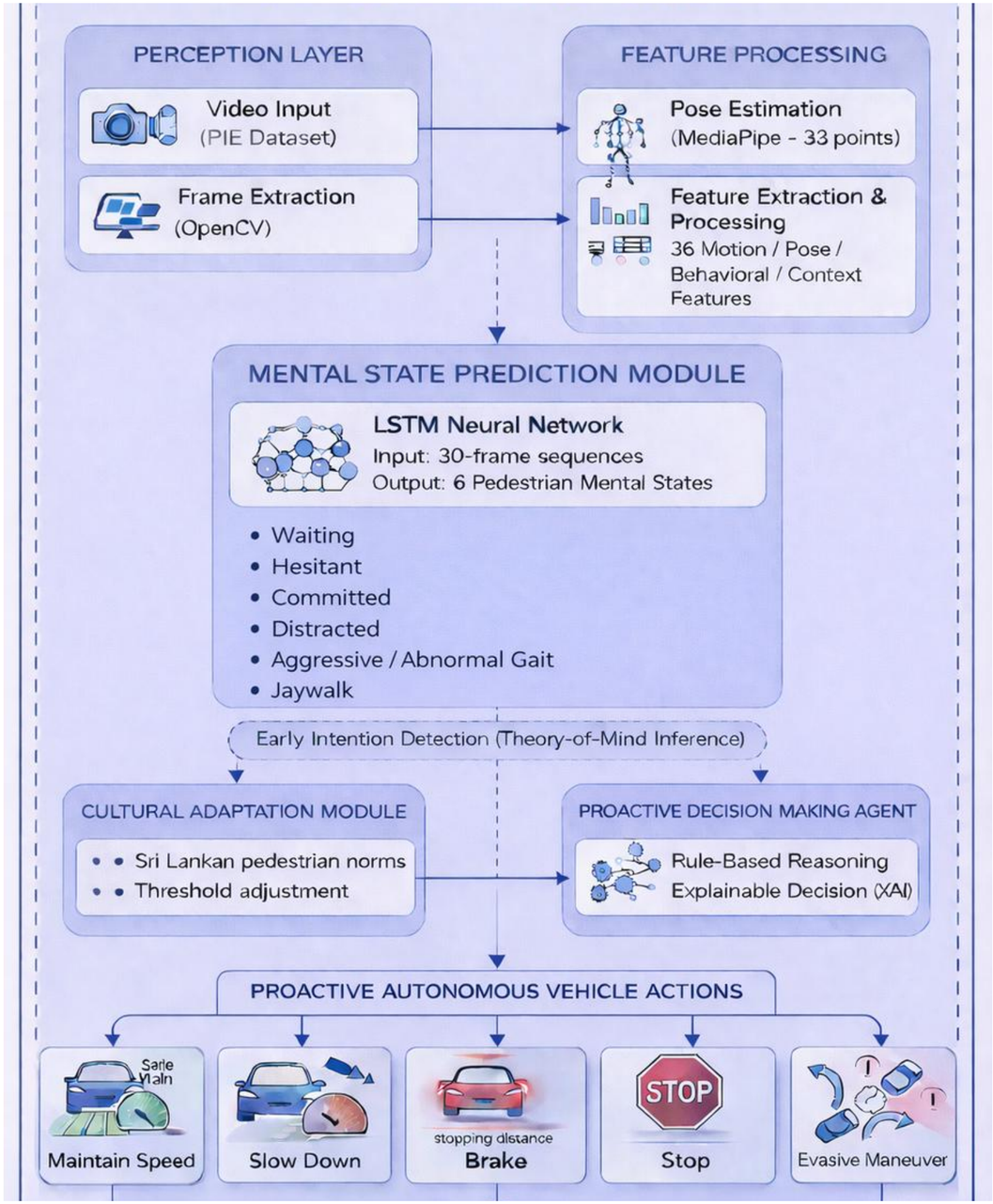
- PIE dataset (2019): 53 videos, 1,800+ crossings
- Publicly available for research purposes

Ethical & Security Clearance

- Public, anonymized datasets used
- No personal information collected
- Testing in simulation environments → no real-world safety risk

Proactive Social Compliance Modeling using Agentic Theory of Mind

System Overview





IT22022974 | Karunaarachchi K A I H

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing



Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

1 Problem Definition

- Autonomous vehicles learn only from local data.
- Traditional federated learning shares all updates, including unreliable information
- Vehicles cannot easily benefit from the rare driving experiences of others.
- This results in limited road intelligence and slow adaptation

2 Novelty

- Agentic federated learning enables selective knowledge sharing.
- An RL-based decision agent determines what knowledge should be shared.
- Filtered updates allow faster and more reliable collective learning.

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

✓ Comparison with Existing Systems

Existing Work	Decides What to Share	Filters Unreliable Updates	Learns from Collective Experience	Handles Rare / Unseen Scenarios
MOSAIC (Nath et al., 2025)	✓ (Similarity-based)	✗	✓ (Agentic)	✗
FedPylot (Quéméneur & Cherkaoui, 2024)	✗ (all updates)	✗	✓ (Federated)	✗
FLAD (Xiang et al., 2025)	✗ (all Updates)	✗	✓ (Federated)	✗
Proposed System	✓ (RL-based)	✓ (Global verification)	✓ (Selective)	✓ (Novelty detection)

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

3 Research Objective

Design and validate an agentic knowledge-sharing framework that enables vehicles to intelligently share knowledge from rare or new road scenarios.

4 Sub-objectives

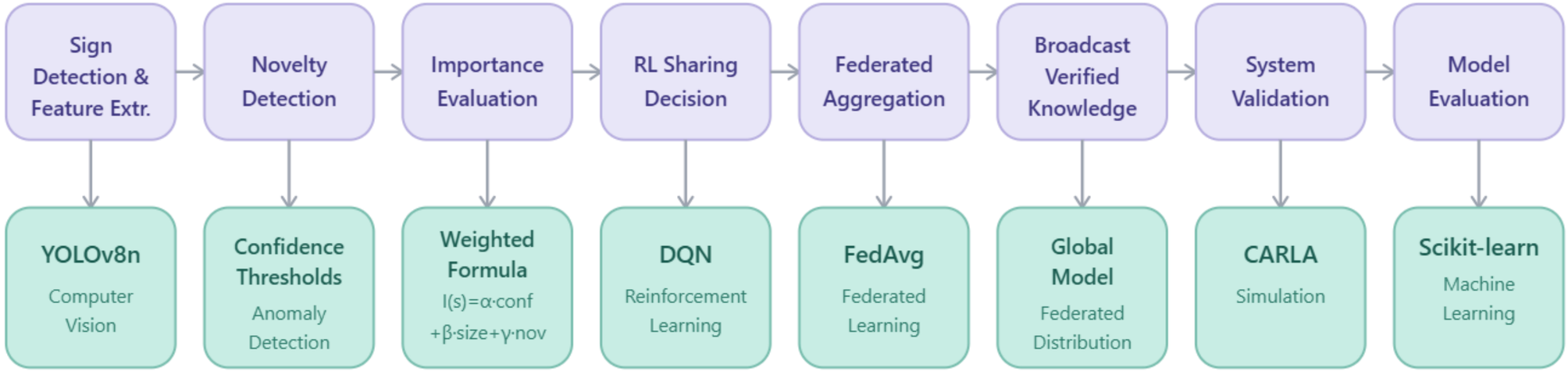
- Develop a traffic sign detection model.
- Develop a method to detect new or rare scenarios
- Develop an importance scoring mechanism
- Implement an intelligent knowledge-sharing agent.
- Develop a verification mechanism for shared knowledge.
- Implement knowledge broadcasting and local model updates.

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

✓ Application of Domain Knowledge

- Autonomous Vehicles
- Artificial Intelligence
- Edge-based Collective Learning

✓ Research Methodology



Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

Validation & Evaluation

- Detection performance is evaluated using precision and recall.
- Sharing efficiency shows model improvement.
- Fleet adaptation measures learning speed for rare or new signs.
- Qualitative evaluation by Domain Experts (AV researchers)
- Simulation-based testing

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

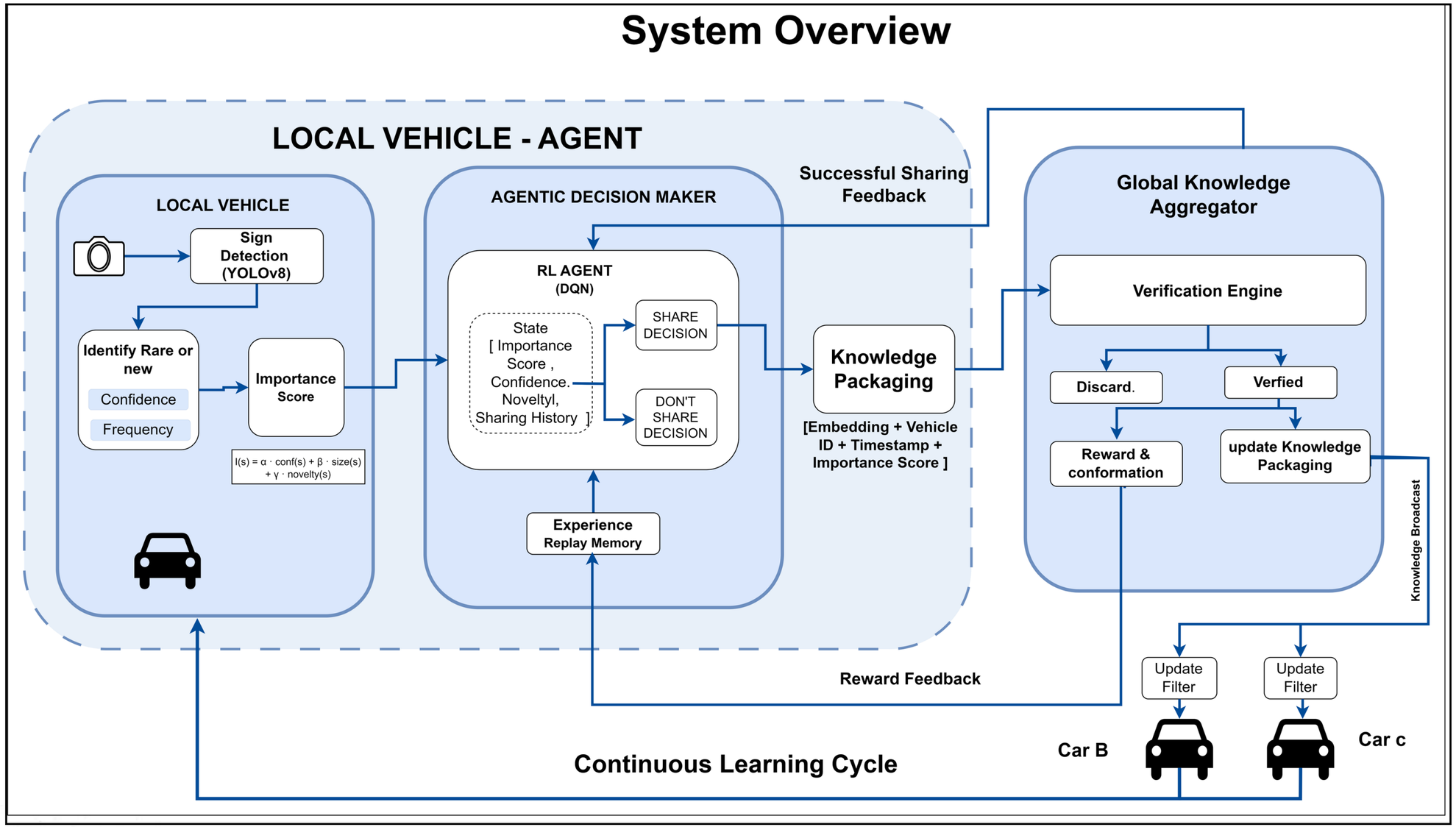
✓ Data Availability

- Dataset: German Traffic Sign Detection Benchmark (GTSDB)
- Images: 900 images (600 training, 300 testing)
- Categories: Prohibitory, Danger, Mandatory

✓ Security & Ethical Clearance

- No personal data is used.
- Only model updates are shared; raw images remain local.
- Public, anonymized datasets ensure ethical compliance.

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing



Agentic AI for Robust and Socially-Aware Autonomous Driving

✓ Commercialization Potential

- Autonomous vehicle market expected to exceed $2 trillion by 2030 (Statista forecast).

- One System, Four Breakthroughs

  See the unseen - Predict pedestrians before they're fully visible

  Negotiate, don't just react - Vehicles resolve right-of-way without infrastructure

  Understand intent, not just movement - ToM reasoning for safer decisions

  Learn as a fleet - Rare knowledge shared selectively, privacy preserved

Agentic AI for Robust and Socially-Aware Autonomous Driving

Investment / Development Cost

Labour Cost – LKR 2.88M

- For developing AI models and system components

Infrastructure Cost – LKR 0.4M

- GPU cloud training (AWS / Google Colab / Azure)
- Data storage and processing

Total Estimated Development Cost ≈ LKR 3.28 M

Pricing Model

- Software licensing to AV manufacturers
  LKR 3,000 – 6,000 per vehicle
- Fleet subscription service for mobility Companies

  Example:
  100 vehicles × LKR 4,500
  ≈ LKR 450,000 revenue

Agentic AI for Robust and Socially-Aware Autonomous Driving

✓ Target Market & Intellectual Property

Target Market / Customers

- Autonomous vehicle manufacturers
- ADAS technology providers
- Smart mobility companies

Examples: Tesla, Waymo, NVIDIA, Cruise

Intellectual Property (IP)

- Patent potential for Agentic AI driving architecture

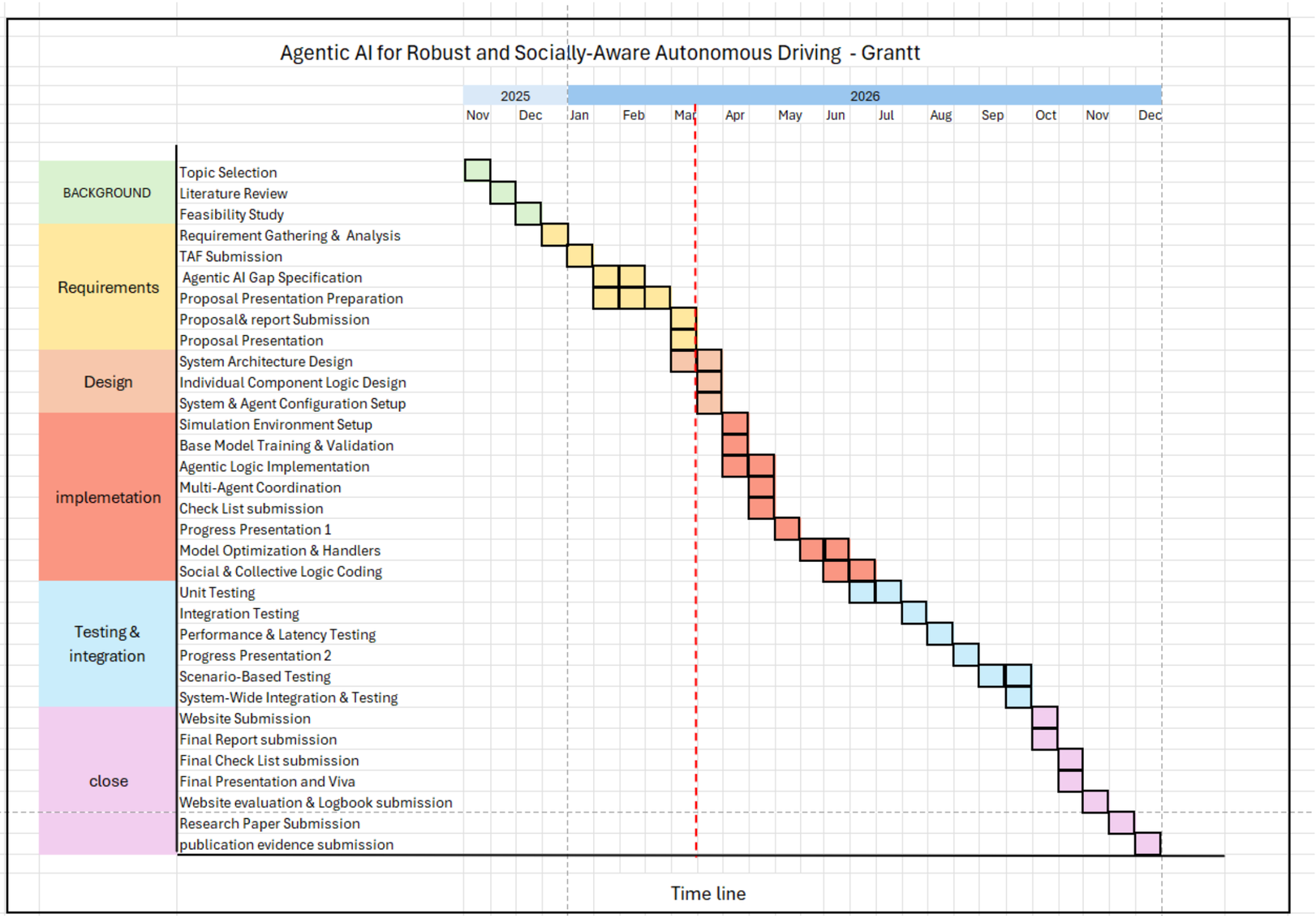








Agentic AI for Robust and Socially-Aware Autonomous Driving



AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING

Work breakdown structure

Work breakdown	Student ID No	Student Name	Research Component
• Occlusion-aware reasoning for hidden pedestrians • Use context & partial observations • Anticipate crossing before full visibility • Early detection of pedestrian risks	IT22245960	Bandara W A P S	Adaptive Intent Prediction in Occluded Urban Scenarios
• Design multi-agent negotiation protocols • Integrate social behavior profiles • Train agents using MARL • Evaluate in SUMO simulation	IT22265906	Nethmini R A I	Multi-Agent Negotiation for Right-of-Way in Complex Intersections
• Extract 24			

Thank You.